

Xinru (Raya) Cheng

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EDUCATION

Peking University

Beijing, China

Master of Software Engineering

Sep 2024 – Jul 2027

- Major Courses: Algorithm analysis and design, Game engines and game development, Natural language processing, Machine learning, Artificial intelligence practices, Software implementation technology, etc.
- Rank: 24/279 (Top 10%)

Beijing University of Chemical Technology

Beijing, China

Bachelor of Digital Media Art

Sep 2020 – Jul 2024

- Major Courses: Programming Foundation, Game Planning and Development, Advanced Mathematics, Advanced Practicum, Virtual Reality Technology, Three Dimensional Animation Design and Production, Graphic Image Processing, Practicum in Applied Software, Graphics of Engineering, etc.
- Rank: 1/30 (Direct Admission to Peking University)

RESEARCH INTERESTS

- Virtual, Augmented, and Mixed Reality Technologies
- Accessibility and Assistive Technologies
- Human-AI Interaction and Next-Generation Interaction
- Computer Vision

ACADEMIC PAPERS

Full Papers

- **Cheng, X.**, Zhang, Y., Wei, Y., Jiang, R., Hansen, P., Liang, H. N. Effects of Spatial Constraint on Bare-Hand Logographic Character Handwriting Performance in Virtual Reality. In *IEEE International Symposium on Mixed and Augmented Reality (ISMAR'26)* [C] & *IEEE Transactions on Visualization and Computer Graphics (TVCG)* [J] (Accepted).
- Wei, Y., **Cheng, X.**, Shi, R., Liang, H. N. Reorienting with the Bare Hand: Gesture-Based Techniques for Orientation-Enabled Teleportation in Virtual Reality. In *IEEE Conference on Virtual Reality and 3D User Interfaces (VR'26)* [C] & *IEEE Transactions on Visualization and Computer Graphics (TVCG)* [J].
- Gao, Y., **Cheng, X.**, Li, Y., Xu, D., Sun, D., Hu, Y. ISDG-Net: Efficient RGB-Infrared Object Detection for Remote Sensing Imagery. In *Remote Sensing* [J].
- Zhang, Y., **Cheng, X.**, Yan, J. AI-Echo: Exploring Technological Alienation Through an Interactive Art Experience. In *International Conference on Human-Computer Interaction (HCII'25)* [C].
- Jiang, R., Zhang, S., Zhao, Y., **Cheng, X.**. The Application of Serious Games in Inclusive Education. In *Game Industry Blue Book: China Game Industry Development Report (2025)*. Beijing: Social Sciences Academic Press [BC].

Posters

- Zhang, Z., Zhang, Y., **Cheng, X.***. Private Non-Visual Addressivity Cues for Blind and Low Vision Users in Mixed-Ability Social VR. In *Conference on Computer-Supported Cooperative Work (CSCW'26)* [C] (Accepted).
- **Cheng, X.**, Zhang, Y., Qiao, Y., Hansen, P., Jiang, R. Negative Danmaku as Social Warning Cues in Video Credibility Assessment. In *European Conference on Computer-Supported Cooperative Work (ECSCW'26)* [C].

Manuscripts Under Review

- Zhang, Y., **Cheng, X.**, Zhao, P., Jiang, R. Exploring the Efficacy of Virtual Reality in Enhancing Human Spatial-Visual Abilities: A Pilot Study on the Spatial Vision VR Training System. In *Games for Health Journal* [J] (Under Third Review).
- Zhang, Y.[†], **Cheng, X.**[†], Liu, R., Feng, X., Chen, B., & Guo, Y. Helping Students Understand Marine in an Immersive Virtual Environment: A Study of the Mechanisms by Which Sense of Presence and Natural Connectedness Enhance Proenvironmental Behavioral Intentions. Submitted to *Ocean & Coastal Management* (Under Review).
- Wang, J., **Cheng, X.**, Hou, B., Liang, H. N. XR-Read: Eye-Level Clarity Matters for Sustained Reading in Head-Mounted Displays. Submitted to *ACM Transactions on Computer-Human Interaction (TOCHI)* (Under Review).

[†] Equal contribution (co-first authors).

* Corresponding author.

PATENTS

- Zhang, P., Li, W., Tang, Z., Wang, Z., Jin, F., **Cheng, X.**.. *An AI Agent System and Collaborative Knowledge Construction Method*. Chinese Invention Patent, ZL 2025 1 1737406.2, Granted May 19, 2026.

RESEARCH EXPERIENCE

Research Assistant, Computational Media and Arts Thrust, Hong Kong University of Science and Technology (Guangzhou) Jul 2025 – Sep 2026

- Supervisor: Hai-Ning Liang
- Finished experimental design, development, and user studies for three parallel projects on VR resolution, reorientation techniques, and handwriting interaction; contributed to three manuscripts accepted/submitted to IEEE TVCG/VR/ISMAR.

Research Intern, Department of Computer Science and Engineering, Hong Kong University of Science and Technology Dec 2025 – Sep 2026

- Supervisor: Xiaojuan Ma
- Participated in the conceptualization and research design of specific tasks such as the human-computer interaction research direction related to speculative artificial products, human-machine collaboration, and the support of large language models.

Visiting Student, Department of Information Management, Peking University Feb 2025 – Jun 2025

- Supervisor: Pengyi Zhang / Preben Hansen
- Led interaction design and front-end development for two AI agent-based educational systems supporting collaborative lesson planning, interactive learning, and knowledge construction; contributed to an NSFC-funded project and a granted Chinese invention patent.

Research Intern, National Virtual Simulation Experimental Teaching Center for Earth Sciences, Peking University Dec 2024 – Jun 2025

- Supervisor: Yanjun Guo
- Conducted research on immersive virtual environments for marine science education, investigating how sense of presence and natural connectedness influence learners' pro-environmental behavioral intentions.

Research Intern, Future Lab - Space and Media Group, Tsinghua University May 2023 – Jul 2023

- Supervisor: Keyang Tang
- Assisted in the development and design of the Digital Luoyang human-computer interaction system.

INDUSTRIAL EXPERIENCE

Investment Intern, Alpha Startups Aug 2025 – Nov 2025

- Conducted pre-investment research, focusing on next-generation interaction technologies and products.
- Participated in all stages of the investment process, including company analysis and market research.

- Research Intern**, MiraclePlus (YC China) Oct 2024 – Nov 2024
- Responsible for communicating with entrepreneurs and planning Demo Day events.
- Product Manager Intern**, Kuaishou Technology Apr 2024 – Aug 2024
- Worked on product development for the e-commerce C-end product department.
- Product Manager Intern**, DiDi Global Nov 2023 – Apr 2024
- Worked on growth product initiatives for the Didi platform, Chitu, and Didi Alliance.
- Interaction Design Intern**, Meituan Jul 2023 – Nov 2023
- Responsible for interaction design and AI capability building in Meituan Cuisine.
 - Received a full-time offer after internship.

ACADEMIC SERVICES

- Teaching Assistant - Game Development Game Engines, Peking University Sep 2024 – Jan 2025
- Teaching Assistant - Artificial Intelligence Practice, Peking University Mar 2025 – Jul 2025
- Reviewer - ISMAR 2025/2026
- Reviewer - CHI 2026
- Reviewer - C&C 2026
- Reviewer - UIST 2026
- Reviewer - CHI PLAY 2026

HONORS & AWARDS

- **National Scholarship (top 1%)**, 2022.
- Shenzhen Stock Exchange Scholarship, 2025 (one of 17 recipients university-wide).
- Outstanding Student Award, Peking University, 2025.
- People's Scholarship (top 5%), 2021,2022,2023.
- Best Student Short Film, Hollywood Independent Film Festival, 2022.
- Third Prize, National College Student Advertising Art Competition, 2022.

SKILLS

- **Programming & Development:** Python, C#, C++, Blueprint, JavaScript, HTML, CSS
- **XR & Creative Technologies:** Unity, Unreal Engine, Processing, TouchDesigner, Kinect, Leap Motion, Arduino
- **3D Modeling & Rendering:** Blender, 3DS MAX, Substance Painter, Zbrush, Octane Render, Marmoset Toolbag, PBR Material Workflow
- **Design & Media Software:** Figma, Adobe Photoshop, After Effects, Animate, Premiere Pro, Illustrator
- **Statistical Analysis:** Descriptive Statistics, Hypothesis Testing (t-test, Chi-Square Test, ANOVA), Regression Analysis, Non-Parametric Tests
- **Data Analysis:** SPSS, Python (pandas, statsmodels), Excel
- **Hobbies:** Pilates, Diving, FPS Games, Piano(Grade 10 of Wuhan Conservatory of Music)